

Little Red Dots: Stellar Eruptions on a Galactic Scale?

Super-Eddington winds mimic black hole signatures in JWST's compact red sources.

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Background

JWST's Little Red Dots are compact, red sources with broad emission lines. Their nature is debated, with some suggesting overmassive black holes.

Their unique spectra don't match any known AGN or galaxy class, hinting at new physics.

What this paper is selling

This paper proposes that LRDs are powered by super-Eddington winds from intermediate-mass engines, similar to eta Carinae's eruption. These winds create a pseudo-photosphere that mimics black hole signatures.

Key results to remember

- LRDs' temperatures cluster like Type II_n supernovae and eta Carinae.
- Broad lines arise from electron scattering in a dense wind, not virial motion.
- Engine masses are likely 10^3 -6 solar masses, not overmassive black holes.

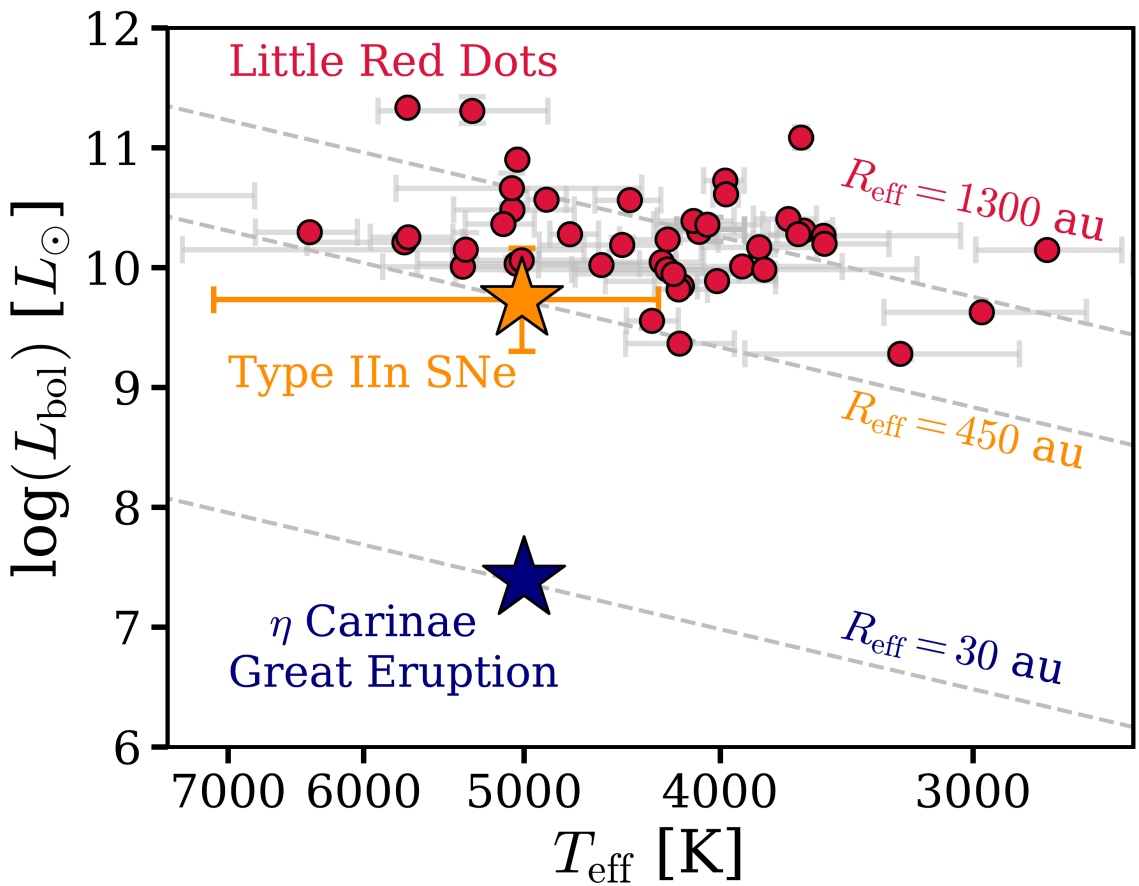


Fig. 4. LRDs, II_n SNe, and eta Carinae share a narrow temperature range despite huge luminosity spread.

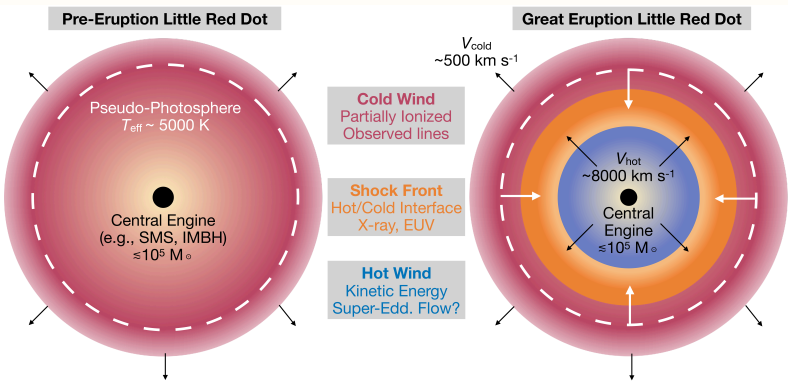


Fig. 8. Schematic: a buried engine drives a cold wind, creating a pseudo-photosphere and shock-heated hot wind.